

Q3 Identity Capacity Plan

Q3 2026 planning | Owner: Mira Chen, Identity SRE | Next review: 2026-08-10

Decision at a glance

Decision

Increase cache-node memory by 25% and schedule directory indexing before August mid-month. Validation capacity is sufficient only if eviction stays low; adding workers cannot compensate for directory tail latency.

Planning peak

3,080 validations/s

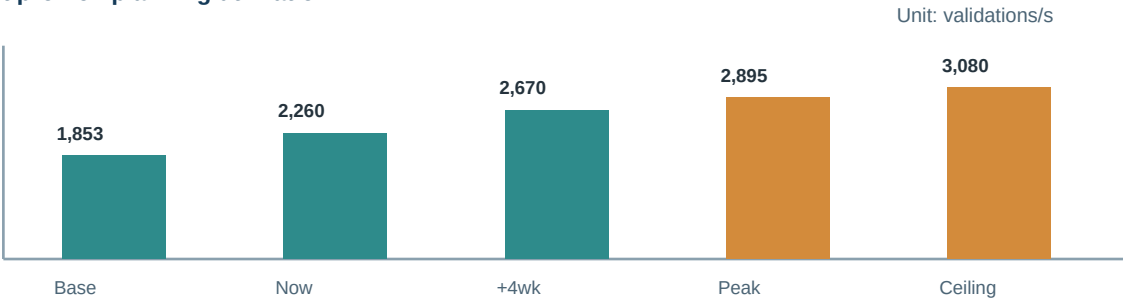
Evidence

High; cache and directory measurements reconciled

Scope and assumptions

Identity planning covers token validation, session refresh, and directory lookup fan-out. The plan assumes a 25% cache-memory expansion, two tenant-style configuration rollouts, and no simultaneous directory reindex. It excludes interactive login rendering because that workload lands outside this service boundary.

Forecast profile - planning utilization



Capacity signals

Measure	Current	Planning limit	Decision signal
Validation rate	2,260 validations/s	3,080 validations/s	Add memory before sustained rate reaches 2,770.
Directory lookup p99	48 ms	< 90 ms	Stop reindex work if lookup tail doubles.
Session-cache eviction	1.1%	< 2.0%	Evictions determine whether extra memory is effective.
Signing key reads	310/s	< 520/s	Unexpected reads indicate cache bypass.

Plan, controls, and ownership

Action plan

Action	Due	Owner	Acceptance evidence
Memory uplift	Jul 28	Identity platform	Increase cache-node memory by 25%.
Index rollout	Aug 05	Directory team	Apply index changes outside the refresh peak.
Bypass probe	Aug 07	Mira Chen	Inject synthetic miss traffic and observe key reads.
Plan review	Aug 10	Security liaison	Approve Q3 envelope with directory owner.

Risk register

Risk	Likelihood	Impact	Response
Directory reindex overlap	Medium	High	Use a maintenance exclusion window for refresh traffic.
Cache bypass defect	Low	High	Alert on signing-key read acceleration.
Key rotation burst	Medium	Medium	Stage rotations across two calendar days.

Decision rationale

Increase cache-node memory by 25% and schedule directory indexing before August mid-month. Validation capacity is sufficient only if eviction stays low; adding workers cannot compensate for directory tail latency.

Selective cross-references

Review alongside: Observability retention capacity for identity trace sampling. These linked plans are relevant because they share a limiting dependency or coordinated operating window.

Model notes: all values are synthetic planning figures. No customer records, production identifiers, secrets, or routable network addresses are included. Northstar Transit Cloud | Identity control-plane planning